「Summary

Bézout

Maximal <=> Prime <=> Irreducible.

Bezout

I.D I.D

$x \in \mathbb{Z}[x,y]$

$2 \in \mathbb{Z}[x]$ $2 \in \mathbb{Z}[\sqrt{-5}]$

Maximal => Prime => Irreducible.

Bézout domain

Noether domain => Atomic domain.

If in=prime.

Field => Euclidean Domain => Principal Ideal Domain => Unique Factorization Domain c Integral Domain.

P.I.D

Maximal <=> Prime <=> Irreducible.

P.I.D.

I.D I.D

U.F.D

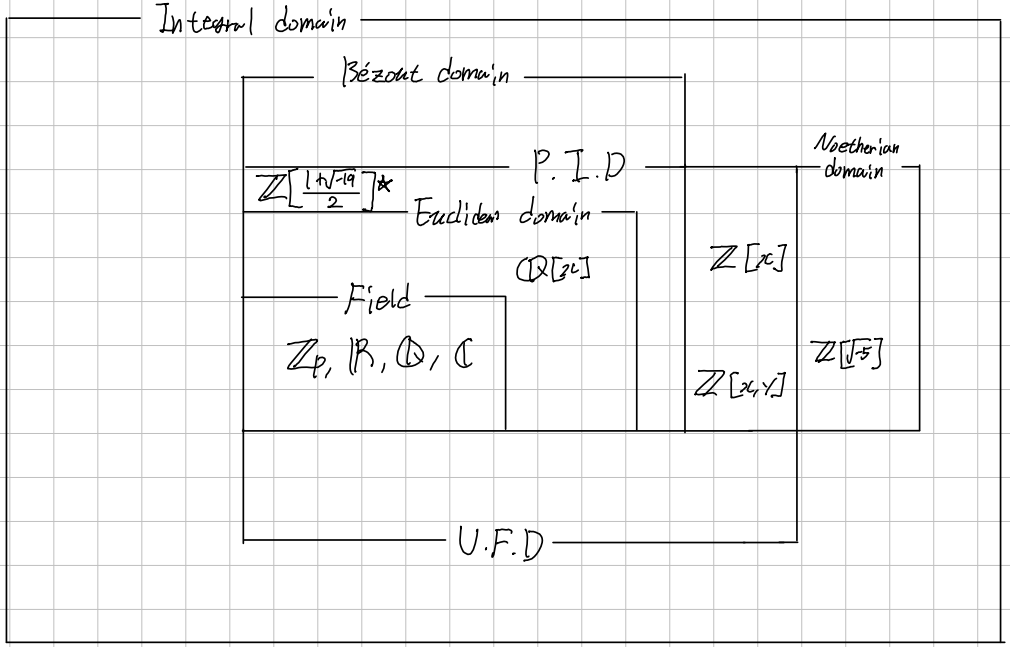
$2 \in \mathbb{Z}[x]$

U.F.D.

Maximal => Prime <=> Irreducible.

P.I.D $\rightarrow$ Noether <=> Every Ideal are finitely generated <=> A.C.C.1 => A.C.C.P => $\exists$ factorization $\rightarrow$ U.F.D

P.I.D $\rightarrow$ Bézout <=> Finitely generated ideal is principal => irreducible=Prime => ! factorization $\rightarrow$ U.F.D



[Basic languages for generated ideal.

[Def. Let $R$ be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1. $a$ is called a multiple of $b$ if there exists $x \in R$ s.t $a = bx$. Simply, $b \mid a$.
2. A non-zero $d \in R$ is called a greatest Common divisor of $a$ and $b$ if;

$d \mid a$ and $d \mid b$ (or, $a, b \in (d)$), $\quad d' \mid a$ and $d' \mid b \Rightarrow d' \mid d$.
$(a, b \in (d') \Rightarrow (d) \subseteq (d'))$

Note: $b \mid a \iff a \in (b) \iff (a) \subseteq (b)$.

Directly, $(a, b) = (d)$ implies $d$ is the g.c.d of $a$ and $b$,
but Conversely, gcd maybe does not generate the ideal; in $\mathbb{Z}[x]$,

$(1) \neq (2, x)$ but $1$ is gcd of $2$ and $x$

[Prop. Let $R$ be a Commutative Ring, and $a, b, c \in R$ be given. Then,

1). $(a, b) = (a) + (b)$
2). $(c) \cdot [(a) + (b)] = (c) \cdot (a) + (c) \cdot (b) = (ca) + (cb)$.
3).